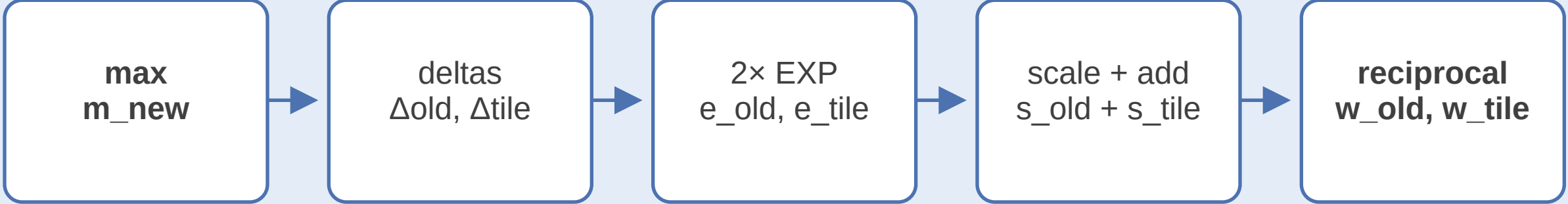


Online Softmax Merge

one merge row

State-dependent scalar recurrence

dependency-bound



loop-carried state • maximum / exponential / reciprocal • low data-level parallelism

specialization target

Regular O[D] vector update

data-parallel

$$O_new[j] = w_old \ O_old[j] + w_tile \ O_tile[j]$$

j=0: independent multiply-add

j=1: independent multiply-add

j=2: independent multiply-add

j=3: independent multiply-add

reuse existing vector datapath